

Eberhard Karls Universität Tübingen

Wirtschafts- und Sozialwissenschaftliche Fakultät

Lehrstuhl für Finance

Prof. Dr. Monika Gehde-Trapp, M.Sc. Tom Ernst

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The Incremental Value of Deep Quantile Networks for Value-at-Risk

Do Neural Networks Add Value to a Realistic Volatility Engine?

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Moritz Maidl

6952271

moritz.maidl@student.uni-tuebingen.de

Abstract

This study examines the incremental value of a Deep Quantile Network (DQN) for Value-at-Risk (VaR) forecasting when a realistic volatility engine is already in place. Using daily EURO STOXX 50 returns, an EGARCH(1,1) model with skew-t innovations is estimated on a long pre-test window and then held fixed to generate a one-dimensional, standardized log volatility state. Conditional on this state, three one-day VaR engines for the 1%, 2.5% and 5% levels are compared: (i) a semi-parametric EGARCH VaR benchmark, (ii) a linear quantile regression and (iii) a deliberately compact DQN. The assessment combines standard VaR backtests, forecast accuracy tests, and evaluations of the economic plausibility of each mapping. Empirically, the benchmark proves superior, delivering robust coverage with hit rates close to nominal targets and near-lowest pinball losses across quantiles. LQR is unable to improve upon this baseline, exhibiting significantly worse forecast accuracy and failing dynamic quantile tests. The DQN fails to add incremental value as it yields generally higher pinball losses than the benchmark and exhibits coverage distortions across different alpha levels. Despite occasionally being conservative in frequency, it exhibits a less stable tail-risk profile consistent with a mapping from volatility to VaR that is irregular and partially non-monotone. Overall, the results indicate that, given a well-specified volatility engine, a compact DQN for VaR estimation does not yield statistical benefits while introducing instability and worsening tail severity behavior.

1 Introduction

VaR remains one of the central internal and regulatory risk metrics for trading books, even though the Basel Committee’s Fundamental Review of the Trading Book and the revised *Minimum capital requirements for market risk* formally shift regulatory requirements to Expected Shortfall (ES) at the 97.5% level (Basel Committee on Banking Supervision, 2013, 2016). In practice, however, many banks and asset managers still run daily VaR engines for monitoring, limits and backtesting, while ES is mainly used in capital calculations as an addition to existing risk measures.

An industry-typical VaR engine first estimates conditional volatility via EWMA or GARCH-type models (including asymmetric variants) with heavy-tailed innovations and then maps volatility into VaR numbers using parametric formulas, filtered historical simulations, or residual quantiles. For major equity indices such as the EURO STOXX 50, an EGARCH(1,1) model with skewed-t innovations is a realistic workhorse specification that captures volatility clustering, leverage effects and heavy tails (Bollerslev, 1986; Engle & Ng, 1993; Hansen, 1994; Nelson, 1991). Recent research proposes non-linear quantile models – random forests, boosted trees and especially DQNs – to forecast VaR more flexibly than linear or parametric specifications (Chronopoulos et al., 2023). While they report sizeable gains

over linear quantile regression and other benchmarks, the adoption of such models in large banks has remained limited. Key concerns include model risk and overfitting in the tails, data-hungry requirements, and interpretability and validation challenges under regulatory model-risk governance, making it difficult to justify a full replacement of established volatility engines with black-box VaR models.

Risk-management desks typically already operate carefully validated volatility engines. The practically relevant question is therefore narrower than in much of the ML literature: conditional on a well-diagnosed volatility engine, is there any tangible benefit in replacing the traditional VaR mapping with a DQN? In Chronopoulos et al. (2023), the DQNs are supplied with high-dimensional predictor sets, effectively subsume both volatility modelling and VaR mapping, and are evaluated primarily on VaR centered metrics, with less emphasis on ES behavior or interpretability. This design is suitable for a forecasting study but less aligned with the incremental, governance-driven changes that risk departments realistically contemplate. From a capital-allocation perspective, ES behavior is at least as important as hit rates. Regulatory reforms explicitly move from VaR to ES to better capture tail risk, highlighting that a model which slightly improves VaR backtests while concentrating risk into fewer but much deeper breaches may be undesirable (Chronopoulos et al., 2023). In addition to that, models have to be economically plausible, transparent and relationships between key drivers like volatility and risk measures should be monotone and interpretable.

The objective of this study is therefore to quantify the incremental value of a compact DQN when plugged on top of a fixed EGARCH(1,1)-skew-t volatility engine for the EURO STOXX 50. From this, three research questions follow:

1. **RQ1:** Given a one-dimensional EGARCH(1,1)-skew-t volatility state, can a compact DQN materially improve one-day VaR forecasts – in terms of hit rates, coverage tests and pinball loss – relative to (i) a semi-parametric EGARCH VaR benchmark and (ii) a linear quantile regression on the same state?
2. **RQ2:** How do the three engines differ with respect to realized shortfall (average loss conditional on VaR breaches) and additional tail-severity metrics (median violation, tail quantiles, worst breaches)? Do any VaR improvements from the DQN come at the cost of deeper and more concentrated tail losses?
3. **RQ3:** Does the DQN’s mapping from volatility to VaR behave monotone and smoothly in the volatility state, or does it display irregular, non-monotone patterns that are difficult to reconcile with standard volatility-risk intuition?

The remainder of the paper is structured as follows. Section 2 introduces the data and the volatility backbone. Section 3 details the methodology and model specifications. Section 4 outlines the evaluation framework, followed by the empirical results in Section 5. Section 6 concludes.

2 Data and Preliminary Analysis

The empirical analysis relies on the Eurozone blue-chip EURO STOXX 50 price index (close-to-close) levels, covering the period from January 1996 to December 2024, totaling 7,449 trading days. Consistent with standard volatility modelling and VaR applications, the price series is converted into daily log-returns. To insure a proper out-of-sample evaluation, the data is split chronologically into a pre-test period (1996–2015) for model estimation and training, and an out-of-sample test period (2016–2024). The pre-test sample is reserved for estimating the EGARCH volatility parameters and training the quantile models (LQR and DQN). The test period is held out for VaR backtesting and economic diagnostics, in order to ensure that no future information contaminates the model selection process.

Table 1 summarizes the distributional properties of the returns. The series exhibits a small positive mean, and returns are centered around zero. The significant negative skewness (-0.23) and excess kurtosis 8.98 (excess 5.98) indicate a distribution that is more peaked and heavy-tailed than a normal density, a finding strongly supported by the rejection of the Jarque–Bera normality test ($p < 0.0001$).

Mean	Standard Deviation	Min	Max	Skewness	Kurtosis	Jarque–Bera
0.016%	1.38%	-13.24%	10.44%	-0.23	8.98	11,152 ($p < 0.0001$)

Table 1: Summary statistics of EURO STOXX 50 daily log-returns (1996–2024)

It is essential to verify that the return series is stationary and that it exhibits conditional heteroscedasticity. Table 2 shows the diagnostic tests. The Augmented Dickey–Fuller (ADF) test strongly rejects the null hypothesis of a unit root ($t = -14.89$), confirming that the log-returns are covariance-stationary. However, the returns are not independent. The ARCH-LM tests on squared returns reject the null of homoscedasticity with extreme significance at lags 5, 10 and 20, providing strong evidence of volatility clustering (Engle, 1982). Furthermore, the Engle–Ng sign and size bias diagnostics ($F = 35.11$) are highly significant, indicating remaining sign/size dependence in conditional variance under a symmetric GARCH baseline (Engle & Ng, 1993).

Test Category	Statistic	p-Value	Conclusion
Stationarity: ADF test on returns	-14.89	< 0.0001	Returns are stationary.
ARCH Effects: ARCH-LM (Lag 10)	1,232.67	< 0.0001	Strong volatility clustering.
Leverage: Engle–Ng Bias Test	35.11	< 0.0001	Asymmetric variance response.

Table 2: Diagnostics for stationarity, heteroscedasticity and leverage effects

These findings – stationary combined with heavy tails, time-varying volatility, and leverage – provide a clear empirical justification for the chosen volatility engine: an EGARCH(1,1) model with skewed Student-t innovations (Bollerslev, 1986; Nelson, 1991).

3 Methodology

In order to quantify the incremental value of neural network-based estimators I employ the previously defined “frozen feature” design. The single volatility model serves as the common backbone for all VaR engines. This ensures that performance differences are attributable solely to the mapping function – whether semi-parametric, linear, or deep neural – rather than differences in the information set (Chronopoulos et al., 2023; J.P. Morgan, 1996).

3.1 EGARCH(1,1)-skew-t volatility engine and volatility state

As decided in the previous section, the conditional volatility process is modeled using an EGARCH(1,1)-skew-t specification to capture volatility clustering, leverage effects, and heavy tails (Hansen, 1994; Nelson, 1991). The return dynamics are defined as $r_t = \mu + \sigma_t z_t$, with innovations $z_t \sim \text{skew-}t(0, 1, \nu, \lambda)$. The log-variance evolves according to

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha g(z_{t-1}) + \gamma z_{t-1}, \quad (1)$$

where $g(\cdot)$ is the EGARCH magnitude term. Parameters are estimated via maximum likelihood on the pre-test period and held fixed during the test period. To serve as the sole input for the LQR and DQN models, I construct a standardized volatility state $x_{\{\sigma, t\}} = (\log \hat{\sigma}_t - \mu_{\log \sigma}) / \sigma_{\log \sigma}$ using the sample moments of the log-volatility observed in the pre-test period; note that $\log \hat{\sigma}_t = \frac{1}{2} \log \hat{\sigma}_t^2$, so this state is a rescaling of the EGARCH log-variance state.

3.2 Benchmark: Semi-parametric EGARCH VaR

The benchmark model combines the parametric volatility dynamics with a non-parametric residual-based tail estimator (Filtered Historical Simulation) (Barone-Adesi et al., 1999). Using the standardized residuals $\hat{z}_t$ from the pre-test period, I compute the empirical α -quantile $\hat{q}_\alpha$ to generate the one-day forecast as

$$\text{VaR}_{\alpha, t}^{\text{EG}} = \hat{\mu} + \hat{\sigma}_t \hat{q}_\alpha. \quad (2)$$

This specification serves as a solid baseline by enforcing the theoretical scaling of risk with volatility while capturing non-Gaussian tail thickness directly from historical data.

3.3 Linear Quantile Regression (LQR)

To test the hypothesis that the mapping from volatility to VaR is linear in log-space, I estimate the specification $Q_\alpha(r_t \mid x_{\{\sigma, t\}}) = \beta_{0, \alpha} + \beta_{1, \alpha} x_{\{\sigma, t\}}$ for each $\alpha \in \{1\%, 2.5\%, 5\%\}$ by minimizing the pinball loss on the pre-test sample (Koenker & Bassett, 1978; Koenker & Hallock, 2001). This model acts as a bridge: if the DQN outperforms the LQR, it implies that non-linearities in the volatility–VaR relationship are economically significant.

3.4 Deep Quantile Network (DQN)

The DQN estimates the conditional quantile function $Q_\alpha(x_{\{\sigma,t\}}; \theta_\alpha)$ using a feed-forward neural network, trained separately for each α to maximize flexibility (Chronopoulos et al., 2023). To mitigate overfitting, the design is deliberately compact, optimized over architectures with $n_L \in \{1, 2\}$ layers and $h \in \{32, 64\}$ using ReLU activations. Network weights are optimized via Adam with L2 weight decay and a derivative-based monotonicity penalty to minimize the empirical pinball (quantile) loss

$$\frac{1}{T} \sum_{t \in \mathcal{T}_{\text{train}}} \rho_\alpha(r_t - Q_\alpha(x_{\{\sigma,t\}}; \theta_\alpha)), \quad (3)$$

where $\rho_\alpha(\cdot)$ denotes the standard quantile (pinball) loss function. Hyperparameters are selected via expanding-window time-series cross-validation on the pre-test data using the unpenalized pinball loss, after which the optimal configuration is retrained on the full pre-test sample.

3.5 Experimental Design and Fairness

To ensure a level playing field, all three VaR engines rely only on the pre-test period for estimation and tuning. The volatility dynamics are held fixed via the EGARCH engine, yielding $\hat{\sigma}_t$. LQR and DQN use $\hat{\sigma}_t$ as their single state input, isolating the effect of the functional form (neural vs. semi-parametric) rather than the predictive power of additional covariates. The semi-parametric benchmark instead combines $\hat{\sigma}_t$ with pre-test residual quantiles. However, this strongly constrains the DQN’s potential.

3.6 Non-Crossing Adjustment

Due to independent estimation, raw VaR forecasts may violate monotonicity (e.g., $\text{VaR}_{1\%,t} > \text{VaR}_{5\%,t}$). To ensure valid ES diagnostics and enforce logical consistency (i.e., $\text{VaR}_{1\%,t} \leq \text{VaR}_{2.5\%,t} \leq \text{VaR}_{5\%,t}$), I apply, for DQN (for others, crossings are negligible), the rearrangement method proposed by Chernozhukov et al. (2010). At each time step t , the vector of raw quantile estimates is sorted ascending:

$$\left(\widehat{\text{VaR}}_{1\%,t}^{\text{adj}}, \widehat{\text{VaR}}_{2.5\%,t}^{\text{adj}}, \widehat{\text{VaR}}_{5\%,t}^{\text{adj}} \right) = \text{sort} \left(\widehat{\text{VaR}}_{1\%,t}, \widehat{\text{VaR}}_{2.5\%,t}, \widehat{\text{VaR}}_{5\%,t} \right). \quad (4)$$

This sorting procedure ensures that the most severe loss prediction is assigned to the 1% level. It preserves the set of predicted tail risks while formally correcting for crossing violations.

4 Evaluation Framework

To ensure a rigorous and holistic evaluation of the competing VaR engines, a multi-layered evaluation framework is employed, moving from standard statistical adequacy tests to forecast accuracy metrics and finally to economic interpretability diagnostics. Specifically, I validate unconditional coverage and independence of violations, rank forecast quality using proper scoring rules, and scrutinize tail severity to ensure that statistical gains do not come at the cost of deeper economic losses.

4.1 Statistical Adequacy and Backtesting

The first layer of assessment focuses on statistical validity using the 2,311 daily out-of-sample observations. To ensure consistency, backtests are applied to the (non-crossing adjusted) VaR series based on the binary violation indicator $I_{t,\alpha} = \mathbb{1}\{r_t < \text{VaR}_{t,\alpha}\}$, which takes the value of 1 if the realized return r_t falls below the forecasted VaR, and 0 otherwise. Unconditional coverage is assessed by comparing the empirical failure rate to the nominal level α and is formally evaluated via the Kupiec likelihood ratio test (Kupiec, 1995),

$$LR_{uc} = -2 \ln \left(\frac{L(\alpha)}{L(\hat{\pi})} \right) \sim \chi_1^2, \quad (5)$$

which tests the null hypothesis $H_0 : E[I_{t,\alpha}] = \alpha$. To account for potential clustering of violations, I additionally apply the Christoffersen conditional coverage test LR_{cc} for joint coverage and independence (Christoffersen, 1998). As a complementary dynamic adequacy check, I employ the Engle–Manganelli Dynamic Quantile (DQ) test (Engle & Manganelli, 2004), which tests whether violations remain unpredictable given lagged violations and the current forecast.

4.2 Forecast Accuracy and Comparison

While backtests establish model adequacy, they do not distinguish between adequate models. To rank forecast quality, I utilize the pinball loss function, a proper scoring rule that balances calibration and sharpness. To determine if performance differences are statistically significant, I apply the Diebold–Mariano (DM) test to the loss differential series $d_t = L_{t,A} - L_{t,B}$ (Diebold & Mariano, 1995). The test statistic is computed as the mean differential $\bar{d}$ divided by its standard error:

$$DM = \frac{\bar{d}}{\sqrt{\widehat{V}(d)/T}} \sim N(0, 1), \quad (6)$$

where $\widehat{V}(d)$ is the long-run variance (HAC) estimate of the loss differential series d_t (Newey & West, 1987).

4.3 Tail Severity and Expected Shortfall Proxies

Since standard VaR backtests ignore the magnitude of breaches, it is critical to ensure that improved coverage does not come at the cost of concentrating risk into deeper tail events. As a proxy for Expected Shortfall, I calculate the Realized Shortfall, defined as the average return conditional on a breach (Acerbi & Székely, 2014):

$$RS_\alpha = \frac{1}{N_{viol}} \sum_{t \in T_{viol}} r_t. \quad (7)$$

A more negative RS_α indicates that the model’s failures are, on average, more severe. To provide a granular view of tail-loss severity beyond the VaR threshold, I also report the median breach return and the 10% quantile (Q10) of the violation magnitude distribution in the test sample.

4.4 Economic Plausibility and Interpretability Diagnostics

In the final step, I assess economic plausibility by analyzing the stability of the volatility-to-VaR mapping. A robust risk model should exhibit a transparent and stable relationship between the input volatility state and the output VaR. I visualize predicted non-crossing adjusted VaR against the standardized volatility state $x_{\{\sigma,t\}}$, shading data-scarce regions (outside the 5% and 95% quantiles) to contextualize extreme behavior. In order to detect local instabilities or lack of convexity, I compute numerical gradients of the non-crossing adjusted VaR mapping to approximate the local sensitivity:

$$\frac{\partial \text{VaR}}{\partial x_{\{\sigma,t\}}} \approx \frac{\text{VaR}(x_{\{\sigma,t\}} + \delta) - \text{VaR}(x_{\{\sigma,t\}} - \delta)}{2\delta}. \quad (8)$$

Economic intuition suggests these should be strictly negative and steepen in high-volatility regimes. Finally, I assess stress-regime behavior by focusing on the highest-volatility bucket (B5) and reporting B5 performance metrics to ensure that aggregate results do not mask systematic failures under crisis conditions.

5 Empirical Results

The empirical analysis evaluates the out-of-sample performance of the three VaR engines over the test period from January 2016 to December 2024. The assessment proceeds from statistical adequacy and forecast accuracy comparisons to a deeper investigation of tail severity and the economic plausibility of the learned mappings.

5.1 Statistical Adequacy and Forecast Accuracy

Model	α	Hit Rate	LRuc (p-val)	LRcc (p-val)	DQ Test (p-val)	Pinball Loss	DM Stat ($H_0: L_{\text{Bench}} = L_{\text{Model } j}$)
Benchmark	1.0%	1.17%	0.428	0.531	0.086	4.12 e-4	—
	2.5%	2.55%	0.871	0.900	0.023	7.89 e-4	—
	5.0%	4.15%	0.055	0.137	0.700	12.81 e-4	—
LQR	1.0%	1.04%	0.853	0.764	0.014	4.25 e-4	$t = -1.00$ ($p = 0.32$)
	2.5%	2.60%	0.768	0.849	0.000	8.18 e-4	$t = -2.65$ ($p = 0.008$)
	5.0%	4.93%	0.882	0.972	0.000	13.24 e-4	$t = -3.74$ ($p < 0.001$)
DQN	1.0%	1.64%	0.004	0.009	0.078	4.28 e-4	$t = -1.71$ ($p = 0.088$)
	2.5%	2.94%	0.185	0.415	0.028	7.86 e-4	$t = 0.44$ ($p = 0.661$)
	5.0%	4.24%	0.086	0.228	0.005	12.96 e-4	$t = -2.36$ ($p = 0.018$)

Table 3: Coverage, Independence, and Forecast Accuracy Metrics

Table 3 indicates that the semi-parametric EGARCH benchmark remains the most reliable specification in this comparison. Across all three quantiles, it delivers hit rates close to the nominal targets and passes both Kupiec and Christoffersen coverage tests, although the DQ test rejects at the 2.5% level ($p = 0.023$), indicating mild residual dynamic misspecification at this level. Notably, the benchmark demonstrates superior predictive accuracy compared to the LQR, which, despite accurate hit rates, consistently fails the DQ test ($p \leq 0.014$) and exhibits statistically significant underperformance in DM tests at the 2.5% ($p = 0.008$) and 5% ($p < 0.001$) levels. The DQN, in contrast, exhibits instability at the far tail, rejecting coverage tests at 1% (1.64% hit rate), though it achieves acceptable coverage at 2.5% and 5%. However, it fails dynamic adequacy tests at the 2.5% and 5% levels. While the DQN achieves a slightly lower, though insignificant, loss at 2.5% ($t = 0.44$), it yields higher pinball losses than the benchmark at the extremes. The DM test confirms statistically significant underperformance specifically at the 5% quantile ($p = 0.018$) and marginal underperformance at 1% ($p = 0.088$). Overall, the joint evidence from coverage, dynamic adequacy, and loss metrics suggests that the DQN mapping does not provide incremental forecasting gains over a well-specified EGARCH volatility backbone.

5.2 Tail Severity and Crisis Behavior

Model	α	Realized Shortfall (RS)	Median Violation	Violation Q10	Crisis Bucket (B5) Hit Rate	Crisis Bucket (B5) RS
Benchmark	1.0%	-3.69%	-3.03%	-6.45%	1.00%	-11.03%
	2.5%	-2.97%	-2.49%	-4.31%	1.90%	-7.82%
	5.0%	-2.66%	-2.14%	-4.04%	4.90%	-5.43%
LQR	1.0%	-3.86%	-3.18%	-7.64%	1.00%	-11.03%
	2.5%	-2.70%	-2.05%	-4.27%	1.90%	-7.82%
	5.0%	-2.35%	-1.94%	-4.06%	6.30%	-5.34%
DQN	1.0%	-3.28%	-2.64%	-4.70%	1.00%	-11.03%
	2.5%	-2.82%	-2.22%	-4.10%	1.50%	-8.73%
	5.0%	-2.48%	-1.98%	-4.02%	3.40%	-6.20%

Table 4: Tail Loss Severity and Crisis-Regime Performance

Table 4 shows that, conditional on the coverage and accuracy results discussed above, the DQN does not produce the most adverse conditional loss profile at tighter confidence levels. Instead, its realized shortfall is generally less negative than for the benchmark (e.g., at the 1% level $RS = -3.28\%$ versus -3.69%), and both the median violation return and lower-tail violation-return quantile (Q10) are likewise milder (e.g., -2.64% and -4.70% at 1%). Importantly, this pattern should not be interpreted as superior tail protection, but as a breach-set selection effect: because the DQN generates more ex-

ceedances, the exceedance set includes more moderate loss days, yielding more frequent but milder average breach returns (losses). In that sense, Table 4 is best read jointly with Table 3, since conditional severity alone is not a sufficient ranking criterion. From an ES-oriented, capital-allocation perspective this remains undesirable, as it reflects weaker threshold calibration rather than improved tail-risk efficiency. The regime-specific crisis bucket supports this interpretation: while B5 hit rates are close to nominal overall, the DQN becomes more conservative in exceedance frequency at higher α , yet its B5 conditional loss severity (RS) is not improved and tends to be more negative than the benchmark. Overall, these crisis-regime severity patterns remain consistent with Section 5.1: the flexibility of the DQN mapping does not generate robust incremental value, and any apparent gains in conditional loss depth should be treated with caution, especially when calibration distortions persist in the left tail.

5.3 Economic Plausibility of the Learned Mappings

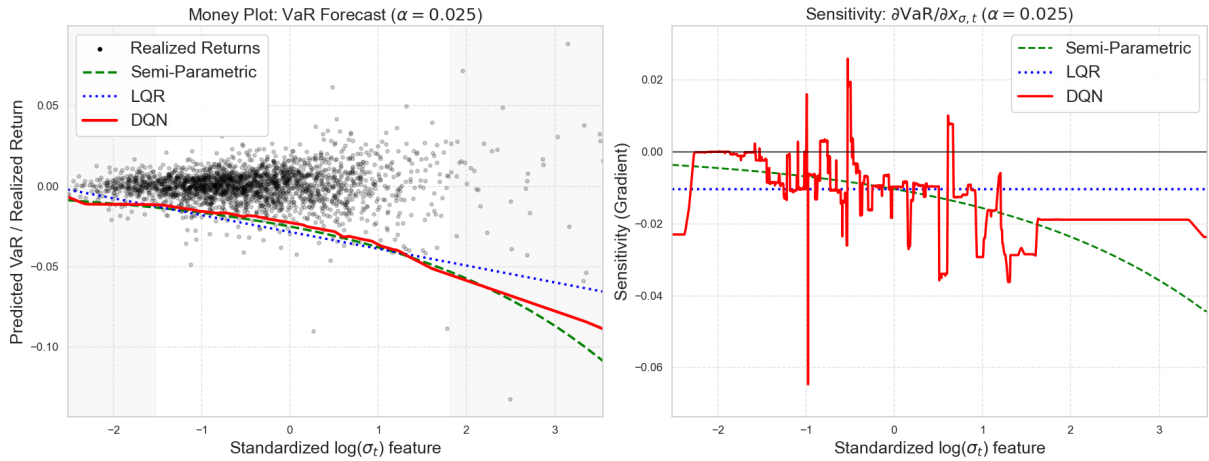


Figure 1: Volatility-to-VaR Mapping Diagnostics at $\alpha = 2.5\%$: Money Plot and Local Sensitivity

The $\alpha = 2.5\%$ (ES-governance-aligned) money plot and corresponding local sensitivity diagnostic provide a check of whether the volatility-to-VaR mapping remains economically plausible: a well-behaved mapping should translate higher conditional volatility into a more negative 2.5% VaR in a smooth, monotone fashion, and the benchmark largely satisfies this expectation by tracing a stable lower envelope of the return cloud, while the LQR provides a linear approximation. The DQN, by contrast, remains the least economically disciplined mapping even after introducing regularization: in the money plot, the monotonicity penalty reduces the most extreme wiggles and local reversals, but the fitted curve still bends erratically in intermediate-to-high volatility regions and becomes visually unreliable where the greyed-out areas indicate pre-test data scarcity. The scatter overlay is informative here: the widening “cone” confirms heteroscedasticity and shows that tail information thins out precisely in the high-volatility range where the DQN must extrapolate. The local sensitivity plot shows that, rather than maintaining a consistently negative slope, the DQN’s estimated derivative continues to jump and inter-

mittently approaches or crosses zero, implying that in some states higher volatility would mechanically map into a less negative 2.5% VaR. This violates the core monotonicity restriction that anchors economic interpretability and suggests that the network is fitting noise. In combination with the backtests and the conditional severity evidence (to be interpreted under breach-set selection when exceedance frequencies differ), the mapping diagnostics therefore indicate that the DQN does not learn a robust VaR boundary, adding functional instability that can yield state-dependent under- or over-conservatism without delivering reliable improvements in tail-risk performance.

6 Discussion and Conclusion

This study wants to answer a practical question: given a realistic and well-specified volatility engine, does replacing only the final volatility-to-VaR mapping with a DQN deliver meaningful incremental value compared with a semi-parametric EGARCH VaR benchmark and a simple linear quantile regression on the same state? The combined evidence provides a consistent answer. Under a fixed one-dimensional EGARCH state, the DQN does not generate robust improvements over the benchmark and is, in economically relevant dimensions, not superior. Any apparent improvement in breach severity reflects selection from differing exceedance rates and does not translate into lower pinball loss across quantiles (1%, 2.5% and 5%), and the learned mapping exhibits irregularities that are difficult to justify under model-risk governance, even under mild regularization. Beyond statistical performance, the mapping diagnostics add an economic perspective. The benchmark and LQR translate higher conditional volatility into a more negative VaR in a stable way. The DQN mapping, in contrast, still exhibits localized flattening and occasional non-monotone responses to volatility, implying that similar volatility states can trigger materially different risk signals. Economically, this suggests that the network is not learning a clearer tail boundary but is instead reshaping risk in a way that weakens the transparency of the volatility–capital link. The study has limitations. Results are based on one major index, one-day-ahead forecasts, and a single volatility specification. The DQN is intentionally restricted to a one-dimensional state and a modest hyperparameter search. However, these constraints are a feature of the research question: they isolate the incremental value of the mapping layer in a realistic governance environment. These conclusions should be read relative to richer Chronopoulos et al. (2023)-type designs, where DQNs add value by exploiting higher-dimensional information sets and alternative horizons. Overall, the study concludes that when a robust EGARCH-skew- t backbone is already in place, the marginal value of a DQN trained solely on the volatility state is limited, and the benchmark remains the more defensible choice for practical risk management. The takeaway is not that DQNs are unhelpful in general, but that under a fixed-engine architecture their incremental value for one-day-ahead EURO STOXX 50 VaR appears small relative to the added interpretability and governance burden.

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